DBMS

ASSIGNMENT 1

QUESTIONS :-

Q1. Write a SQL statement to create a simple table countries including columns country id, country name and region id.

Ans:
```
CREATE TABLE countries (
  Country _id VARCHAR(10),
  Country _name VARCHAR(50),
  Region _id INT
);
```

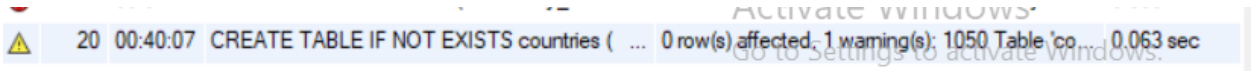
Output :



Q2. Write a SQL statement to create a simple table countries including columns country id, country name and region id which is already exists.

Ans :
```
CREATE TABLE IF NOT EXISTS countries (
  Country _id VARCHAR(10),
  country _name VARCHAR(50),
  region _id INT
);
```

Output :



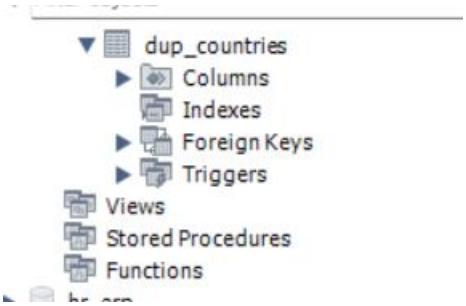
Q3. Write a SQL statement to create the structure of a table dup_countries similar to countries.

ANS: CREATE TABLE dup_countries AS

SELECT * FROM countries

WHERE 1 = 0;

Output:

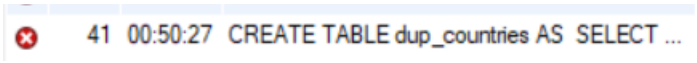


Q4. Write a SQL statement to create a duplicate copy of countries table including structure and data by name dup_countries.

Ans: CREATE TABLE dup_countries AS

SELECT * FROM countries;

Output:



Q5. Write a SQL statement to create a table countrie set a constraint NULL.

Ans: CREATE TABLE countrie (

```
  country_id VARCHAR(10) NULL,
  country_name VARCHAR(50) NULL,
  region_id INT NULL
);
```

Output:

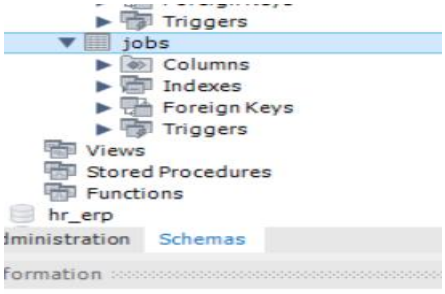


Q6. Write a SQL statement to create a table named jobs including columns job_id, job_title, min_salary, max_salary and check whether the max_salary amount exceeding the upper limit 25000.

Ans;

```
CREATE TABLE jobs (
  job_id VARCHAR(10) PRIMARY KEY,
  job_title VARCHAR(50),
  min_salary DECIMAL(10, 2),
  max_salary DECIMAL(10, 2),
  CONSTRAINT chk_max_salary CHECK (max_salary <= 25000)
);
```

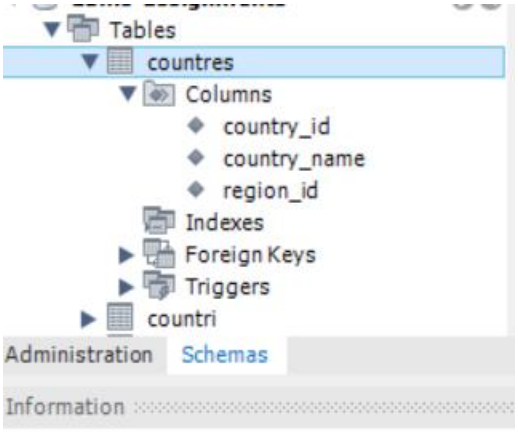
Output:



Q7. Create table countries with a CHECK constraint to allow only Italy, India, and China

```
CREATE TABLE countres (
   country_id VARCHAR(10),
   country_name VARCHAR(50),
   region_id INT,
   CONSTRAINT chk_country CHECK (country_name IN ('Italy', 'India', 'China'))
);
```

OUTPUT:



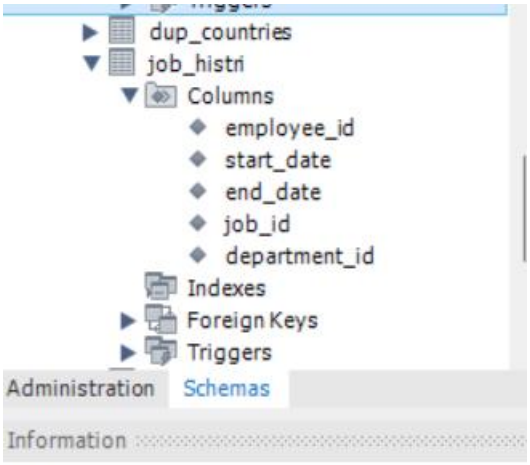


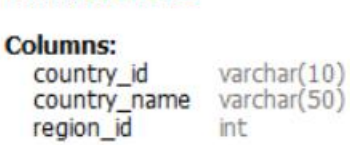
Q8. Create table job_histri with a CHECK constraint for the date format '--/--/----'

Ans:
```
CREATE TABLE job_histri (
```

```
  employee_id INT,
  start_date VARCHAR(20),
  end_date VARCHAR(20),
  job_id VARCHAR(10),
  department_id INT,
  CONSTRAINT chk_end_date CHECK (end_date LIKE '--/--/----')
);
```

Output:



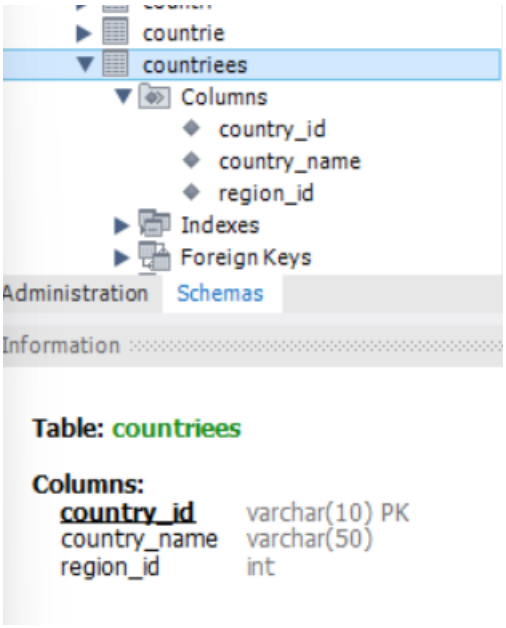


Q9. Create table countries with a PRIMARY KEY / UNIQUE constraint on country_id to prevent duplicates

ANS:
```
CREATE TABLE countriees (
  country_id VARCHAR(10) PRIMARY KEY,
```

```
    country_name VARCHAR(50),
    region_id INT
);
```

Output:

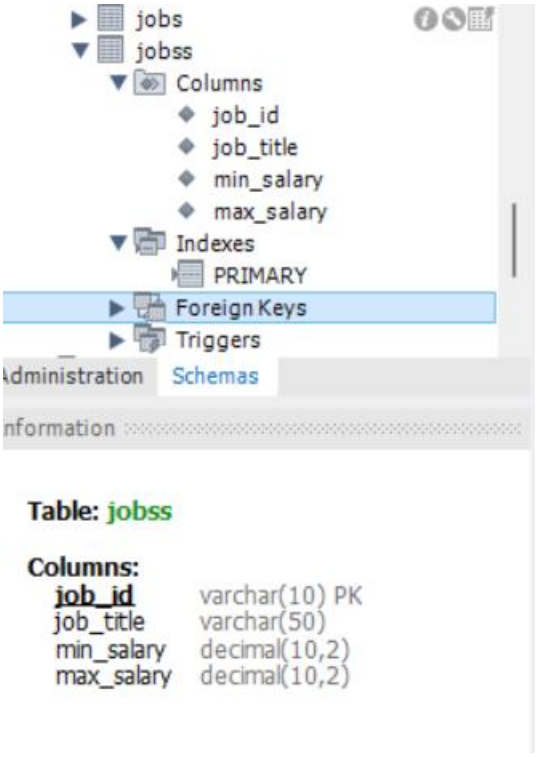


Q10. Create table jobss with DEFAULT constraints for job_title, min_salary, and max_salary

Ans:

```
CREATE TABLE jobss (
    job_id VARCHAR(10) PRIMARY KEY,
    job_title VARCHAR(50) DEFAULT '',
    min_salary DECIMAL(10, 2) DEFAULT 8000,
    max_salary DECIMAL(10, 2) DEFAULT NULL
);
```

Output:



-- 11. Create table countr where country_id is a key field (PRIMARY KEY) preventing duplicates

Ans:

```
CREATE TABLE countr (
  country_id VARCHAR(10) PRIMARY KEY,
  country_name VARCHAR(50),
  region_id INT
);
```

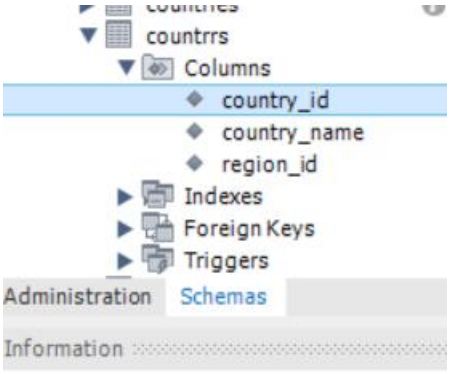
Output:



-- 12. Create table countrrs where country_id is unique and stores an auto-incremented value

```
CREATE TABLE countrrs (
  country_id INT AUTO_INCREMENT PRIMARY KEY,
  country_name VARCHAR(50),
  region_id INT
);
```

Output:



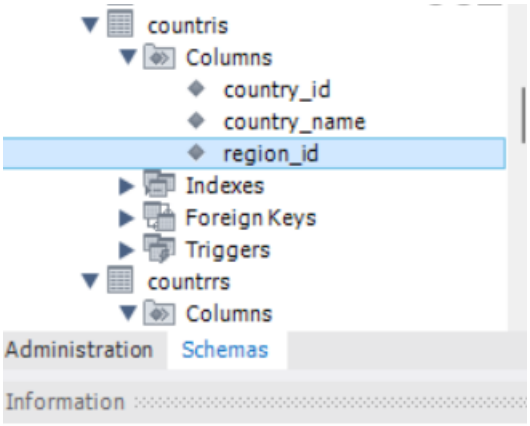
Q13. Create table countris where the combination of country_id and region_id is unique

Ans:

```
CREATE TABLE countris (
  country_id VARCHAR(10),
  country_name VARCHAR(50),
  region_id INT,
  CONSTRAINT unique_country_region UNIQUE (country_id, region_id)
```

);

Output:



Column: region_id

Definition:
region_id int

14. Write a SQL statement to create a table job_history including columns employee_id, start_date, end_date, job_id and department_id and make sure , the employee_id column does not contain any duplicate value at the time of insertion and the foreign key column job_id contain only those values which are exists in the jobs table.

Ans:
```
CREATE TABLE job_history (
  employee_id INT PRIMARY KEY,
  start_date DATE,
  end_date DATE,
  job_id VARCHAR(10),
  department_id INT,
  CONSTRAINT fk_job FOREIGN KEY (job_id) REFERENCES jobs(JOB_ID)
);
```

Output:

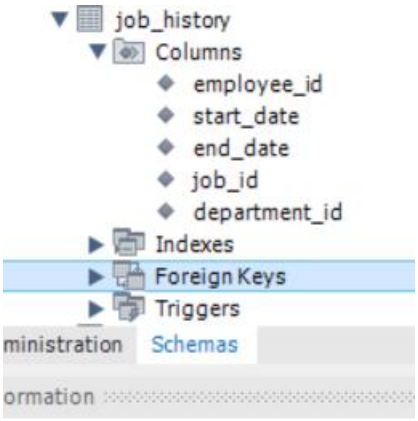


Table: job_history

Columns:

employee_id	int PK
start_date	date
end_date	date
job_id	varchar(10)
department_id	int

15. Write a SQL statement to create a table employees including columns employee_id, first_name, last_name, email, phone_number hire_date, job_id, salary, commission, manager_id and department_id and make sure that, the employee_id column does not contain any duplicate value at the time of insertion and the foreign key columns combined by department_id and manager_id columns contain only those unique combination values, which combinations are exists in the departments table.

Ans:
```
CREATE TABLE employees (
  employee_id INT PRIMARY KEY,
  first_name VARCHAR(50),
  last_name VARCHAR(50),
```

```
    email VARCHAR(100),
    phone_number VARCHAR(20),
    hire_date DATE,
    job_id VARCHAR(10),
    salary DECIMAL(10, 2),
    commission DECIMAL(5, 2),
    manager_id DECIMAL(6, 0),
    department_id DECIMAL(4, 0),
    CONSTRAINT fk_dept_mgr FOREIGN KEY (department_id, manager_id)
    REFERENCES departments(DEPARTMENT_ID, MANAGER_ID)
);
```

Output:

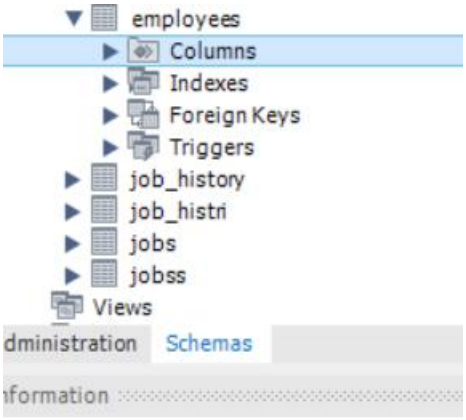


Table: employees

Columns:

employee_id	int PK
first_name	varchar(50)
last_name	varchar(50)
email	varchar(100)
phone_number	varchar(20)
hire_date	date
job_id	varchar(10)
salary	decimal(10,2)
commission	decimal(5,2)
manager_id	decimal(6,0)
department_id	decimal(4,0)

Q16. Write a SQL statement to create a table employees including columns employee_id, first_name, last_name, email, phone_number hire_date, job_id, salary, commission, manager_id and department_id and make sure that, the employee_id column does not contain any duplicate value at the time of insertion, and the foreign key column department_id, reference by the column department_id of departments table, can contain only those values which are exists in the departments table and another foreign key column job_id, referenced by the column job_id of jobs table, can contain only those values which are exists in the jobs table. The InnoDB Engine have been used to create the tables.

Ans:

```
CREATE TABLE employes (
  employee_id INT PRIMARY KEY,
  first_name VARCHAR(50),
  last_name VARCHAR(50),
  email VARCHAR(100),
  phone_number VARCHAR(20),
  hire_date DATE,
  job_id VARCHAR(10),
  salary DECIMAL(10, 2),
  commission DECIMAL(5, 2),
  manager_id DECIMAL(6, 0),
  department_id DECIMAL(4, 0),
  CONSTRAINT fk_job_emp FOREIGN KEY (job_id) REFERENCES jobs(JOB_ID),
```

CONSTRAINT fk_dept_emp FOREIGN KEY (department_id) REFERENCES departments(DEPARTMENT_ID)

) E

O



Table: employes

Columns:

employee_id	int PK
first_name	varchar(50)
last_name	varchar(50)
email	varchar(100)
phone_number	varchar(20)
hire_date	date
job_id	varchar(10)
salary	decimal(10,2)
commission	decimal(5,2)
manager_id	decimal(6,0)
department_id	decimal(4,0)

Q17. Write a SQL statement to create a table employees including columns employee_id, first_name, last_name, job_id, salary and make sure that, the employee_id column does not contain any duplicate value at the time of insertion, and the foreign key column job_id, referenced by the column job_id of jobs table, can contain only those values which are exists in the jobs table. The InnoDB Engine have been used to create the tables. The specialty of the statement is that, The ON UPDATE CASCADE action allows you to perform cross-table update and ON DELETE RESTRICT action reject the deletion. The default action is ON DELETE RESTRICT.

Ans: CREATE TABLE employess (

employee_id INT PRIMARY KEY,

first_name VARCHAR(50),

last_name VARCHAR(50),

job_id INT,

salary DECIMAL(10, 2));

Output:



Q18. Write a SQL statement to create a table employees including columns employee_id, first_name, last_name, job_id, salary and make sure that, the employee_id column does not contain any duplicate value at the time of insertion, and the foreign key column job_id, referenced by the column job_id of jobs table, can contain only those values which are exists in the jobs table. The InnoDB Engine have been used to create the tables. The specialty of the statement is that, The ON DELETE CASCADE that lets you allow to delete records in the employees(child) table that refer to a record in the jobs(parent) table when the record in the parent table is deleted and the ON UPDATE RESTRICT actions reject any updates.

Ans: CREATE TABLE employesss (

employee_id INT PRIMARY KEY,

```
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    job_id INT,
    salary DECIMAL(10, 2),
    CONSTRAINT fk_job_emp FOREIGN KEY (job_id)
    REFERENCES jobs(JOB_ID)
    ON DELETE CASCADE
    ON UPDATE RESTRICT
) ENGINE = InnoDB;
```

C

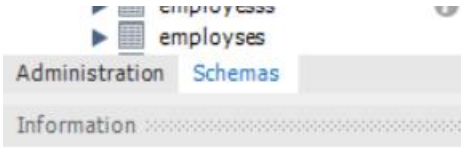


Q19. Write a SQL statement to create a table employees including columns employee_id, first_name, last_name, job_id, salary and make sure that, the employee_id column does not contain any duplicate value at the time of insertion, and the foreign key column job_id, referenced by the column job_id of jobs table, can contain only those values which are exists in the jobs table. The InnoDB Engine have been used to create the tables. The specialty of the statement is that, The ON DELETE SET NULL

action will set the foreign key column values in the child table(employees) to NULL when the record in the parent table(jobs) is deleted, with a condition that the foreign key column in the child table must accept NULL values and the ON UPDATE SET NULL action resets the values in the rows in the child table(employees) to NULL values when the rows in the parent table(jobs) are updated.

Ans:

```
CREATE TABLE employeses (
  employee_id INT PRIMARY KEY,
  first_name VARCHAR(50),
  last_name VARCHAR(50),
  job_id INT NULL,
  salary DECIMAL(10, 2),
  CONSTRAINT fk_job_emp FOREIGN KEY (job_id)
  REFERENCES jobs(JOB_ID)
  ON DELETE SET NULL
  ON UPDATE SET NULL
) ENGINE = InnoDB;
```





Q20. Write a SQL statement to create a table employees including columns employee_id, first_name, last_name, job_id, salary and make sure that, the employee_id column does not contain any duplicate value at the time of insertion, and the foreign key column job_id, referenced by the column job_id of jobs table, can contain only those values which are exists in the jobs table. The InnoDB Engine have been used to create the tables. The specialty of the statement is that, The ON DELETE NO ACTION and the ON UPDATE NO ACTION actions will reject the deletion and any updates.

Ans:

```
CREATE TABLE employesss (
  employee_id INT PRIMARY KEY,
  first_name VARCHAR(50),
  last_name VARCHAR(50),
  job_id INT,
  salary DECIMAL(10, 2),
  CONSTRAINT fk_job_emp FOREIGN KEY (job_id)
  REFERENCES jobs(JOB_ID)
  ON DELETE NO ACTION
  ON UPDATE NO ACTION
) ENGINE = InnoDB;
```

Output:

